KAIZEN MATHS

Lesson starter

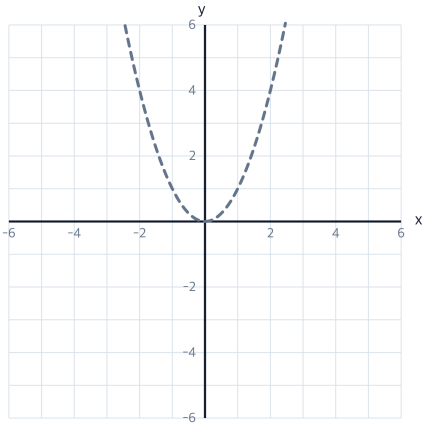
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

SA1

Describe the combined transformation and sketch the image graph.



y = f(x)image to be sketched

The dashed graph is $f(x) = x^2$. Sketch $y = 2f(x + 2) + 3$ and describe the transformations.

Show your working

KAIZEN MATHS

Lesson starter

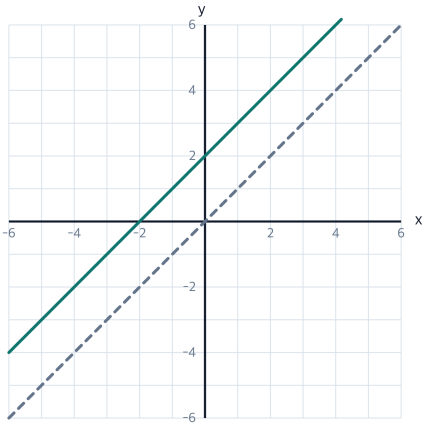
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

SB1

Compare the original graph and image graph, then describe the transformation fully.



dashed originalsolid image

The dashed graph is y = f(x) and the solid graph is the image. Describe the transformation fully.

Show your working

KAIZEN MATHS

Lesson starter

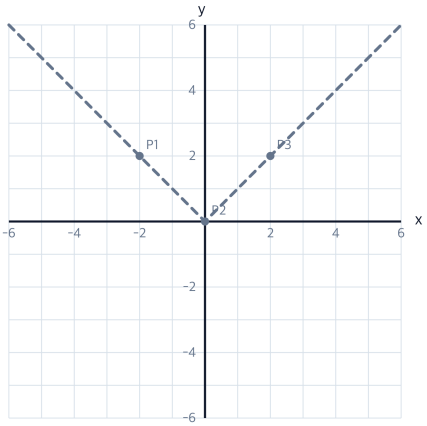
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

SC1

Transform the key points and use them to sketch the image graph.



given graphimage to be sketched

The key points on y = f(x) are (-2, 2), (0, 0), (2, 2). Use y = f(x - 3) to find the image points.

Show your working

KAIZEN MATHS

Lesson starter

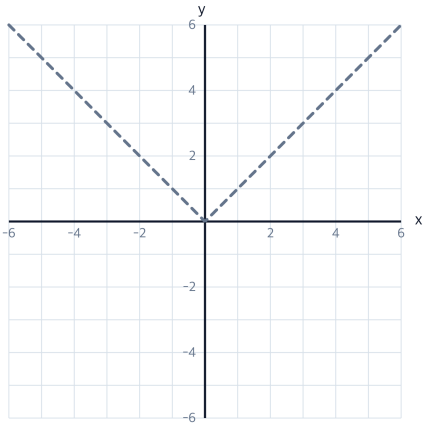
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

SD1

Describe the combined transformation and sketch the image graph.



y = f(x)image to be sketched

The dashed graph is f(x) = |x|. Sketch y = 2f(x + 3) - 2 and describe the transformations.

Show your working

KAIZEN MATHS

Exit ticket

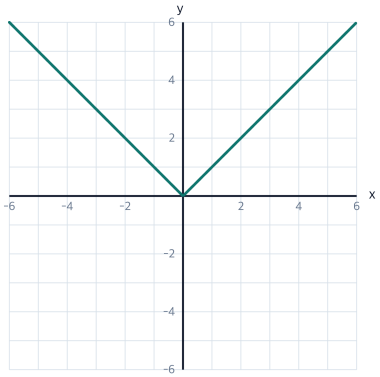
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

EA1

Compare the original graph and image graph, then describe the transformation fully.



dashed originalsolid image

The dashed graph is y = f(x) and the solid graph is the image. Describe the transformation fully.

ALG-146D-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

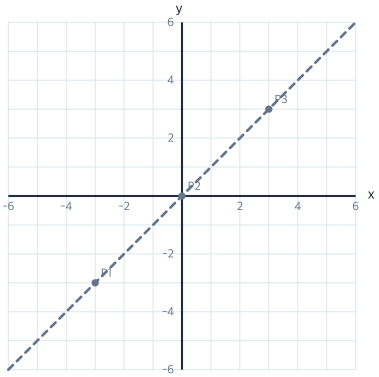
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

EB1

Transform the key points and use them to sketch the image graph.



given graphimage to be sketched

The key points on y = f(x) are (-3, -3), (0, 0), (3, 3). Use y = f(x - 3) to find the image points.

ALG-146D-L3 · EB

Show your working

KAIZEN MATHS

Exit ticket

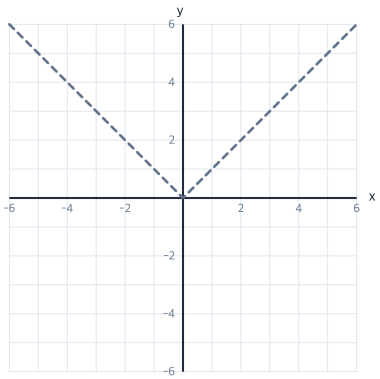
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

EC1

Describe the combined transformation and sketch the image graph.



y = f(x)image to be sketched

The dashed graph is f(x) = |x|. Sketch y = 2f(x - 2) + 3 and describe the transformations.

ALG-146D-L3 · EC

Show your working

KAIZEN MATHS

Exit ticket

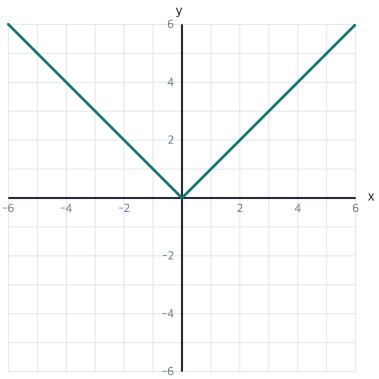
Graph Transformations and Curve Sketching

Combined Transformations

Name _____

EA1

Compare the original graph and image graph, then describe the transformation fully.



dashed originalsolid image

The dashed graph is y = f(x) and the solid graph is the image. Describe the transformation fully.

ALG-146D-L3 · EA

Show your working

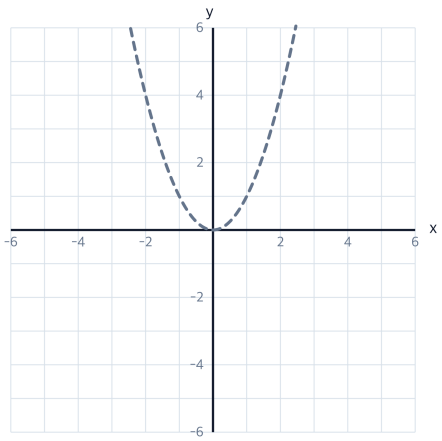
Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

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SA1

Describe the combined transformation and sketch the image graph.



y = f(x)image to be sketched

The dashed graph is $f(x) = x^2$. Sketch $y = 2f(x + 2) + 3$ and describe the transformations.

Answer:

Translate left 2, stretch vertically by scale factor 2, then translate up 3.

Working:

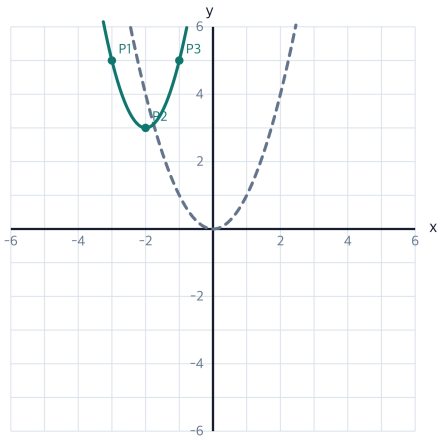
Identify the parent graph: $f(x) = x^2$.

The equation $y = 2f(x + 2) + 3$ uses the point mapping $(x, y) \rightarrow (x - 2, 2y + 3)$.

Map key points from the parent graph.

Original point	Image point
(-2, 4)	(-4, 11)
(-1, 1)	(-3, 5)
(0, 0)	(-2, 3)
(1, 1)	(-1, 5)
(2, 4)	(0, 11)

Sketch the transformed curve through the image points.



Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

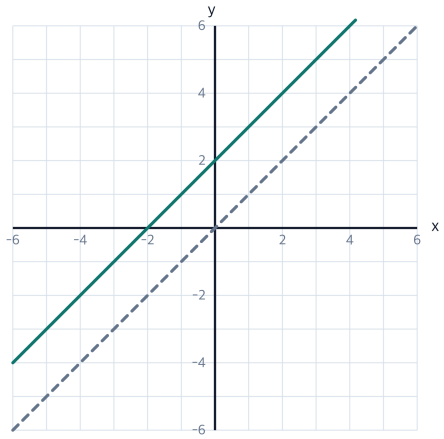
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SA1 (continued)

$y = f(x) y = 2f(x + 2) + 3$

SB1

Compare the original graph and image graph, then describe the transformation fully.



dashed originalsolid image

The dashed graph is y = f(x) and the solid graph is the image. Describe the transformation fully.

Answer:

Translation 2 units up.

Working:

Compare a clear point on the dashed graph with the corresponding point on the solid graph.

The points follow the rule (x, y) → (x, y + 2).

This means the transformation is: Translation 2 units up.

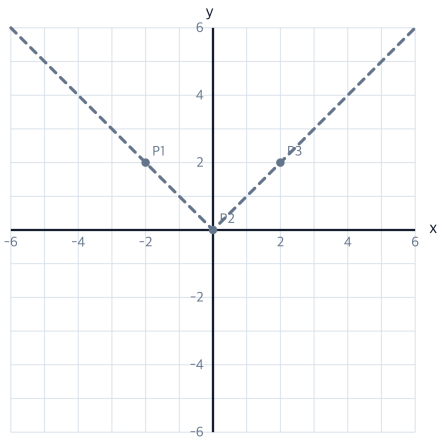
Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

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SC1

Transform the key points and use them to sketch the image graph.



given graphimage to be sketched

The key points on y = f(x) are (-2, 2), (0, 0), (2, 2). Use y = f(x - 3) to find the image points.

Answer:

Image points: (1, 2), (3, 0), (5, 2).

Working:

Use the mapping rule for y = f(x - 3): (x, y) → (x + 3, y).

Apply the rule to each original point.

Original point	Image point
(-2, 2)	(1, 2)
(0, 0)	(3, 0)
(2, 2)	(5, 2)

The image points are (1, 2), (3, 0), (5, 2).

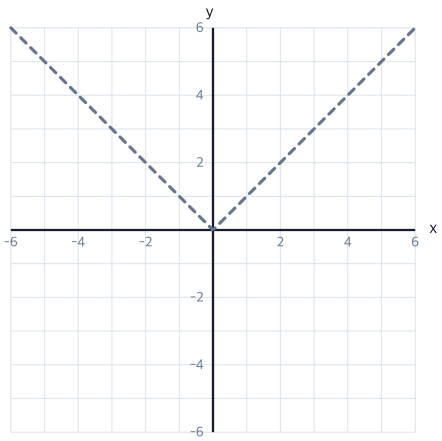
Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

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SD1

Describe the combined transformation and sketch the image graph.



y = f(x)image to be sketched

The dashed graph is f(x) = |x|. Sketch y = 2f(x + 3) - 2 and describe the transformations.

Answer:

Translate left 3, stretch vertically by scale factor 2, then translate down 2.

Working:

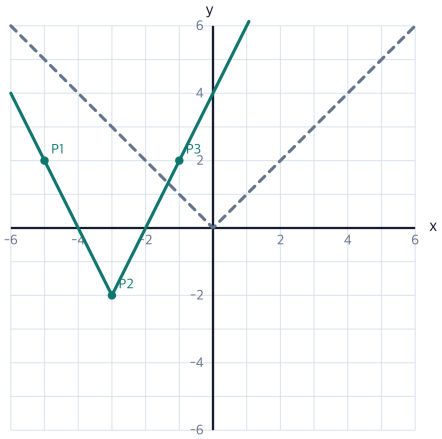
Identify the parent graph: f(x) = |x|.

The equation y = 2f(x + 3) - 2 uses the point mapping (x, y) → (x - 3, 2y - 2).

Map key points from the parent graph.

Original point	Image point
(-2, 2)	(-5, 2)
(0, 0)	(-3, -2)
(2, 2)	(-1, 2)

Sketch the transformed curve through the image points.



y = f(x)y = 2f(x + 3) - 2

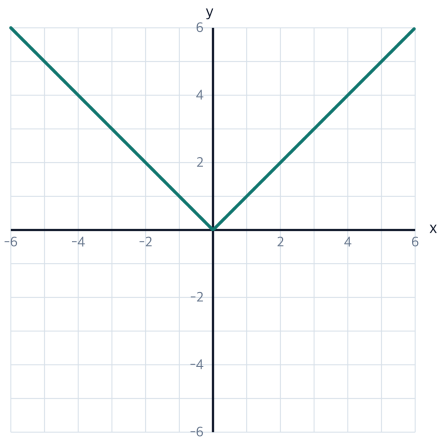
Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

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EA1

Compare the original graph and image graph, then describe the transformation fully.



dashed originalsolid image

The dashed graph is y = f(x) and the solid graph is the image. Describe the transformation fully.

Answer:

Reflection in the y-axis.

Working:

Compare a clear point on the dashed graph with the corresponding point on the solid graph.

The points follow the rule (x, y) → (-x, y).

This means the transformation is: Reflection in the y-axis.

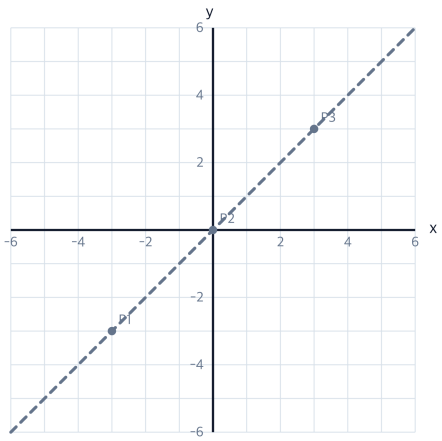
Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

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EB1

Transform the key points and use them to sketch the image graph.



given graphimage to be sketched

The key points on y = f(x) are (-3, -3), (0, 0), (3, 3). Use y = f(x - 3) to find the image points.

Answer:

Image points: (0, -3), (3, 0), (6, 3).

Working:

Use the mapping rule for y = f(x - 3): (x, y) → (x + 3, y).

Apply the rule to each original point.

Original point	Image point
(-3, -3)	(0, -3)
(0, 0)	(3, 0)
(3, 3)	(6, 3)

The image points are (0, -3), (3, 0), (6, 3).

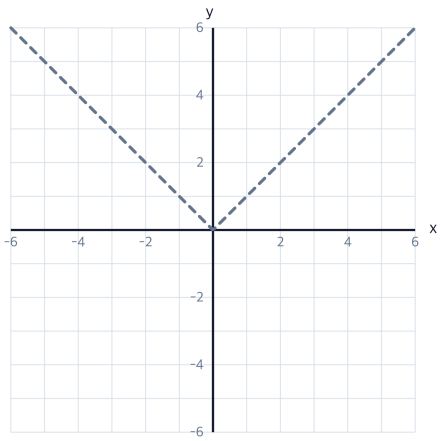
Teacher answers & working

Graph Transformations and Curve Sketching · Combined Transformations

ALG-146D-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Describe the combined transformation and sketch the image graph.



y = f(x)image to be sketched

The dashed graph is f(x) = |x|. Sketch y = 2f(x - 2) + 3 and describe the transformations.

Answer:

Translate right 2, stretch vertically by scale factor 2, then translate up 3.

Working:

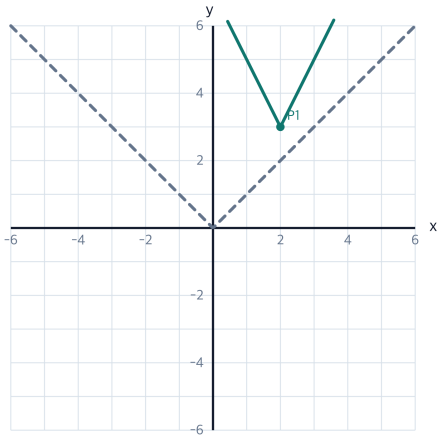
Identify the parent graph: f(x) = |x|.

The equation y = 2f(x - 2) + 3 uses the point mapping (x, y) → (x + 2, 2y + 3).

Map key points from the parent graph.

Original point	Image point
(-2, 2)	(0, 7)
(0, 0)	(2, 3)
(2, 2)	(4, 7)

Sketch the transformed curve through the image points.



y = f(x)y = 2f(x - 2) + 3